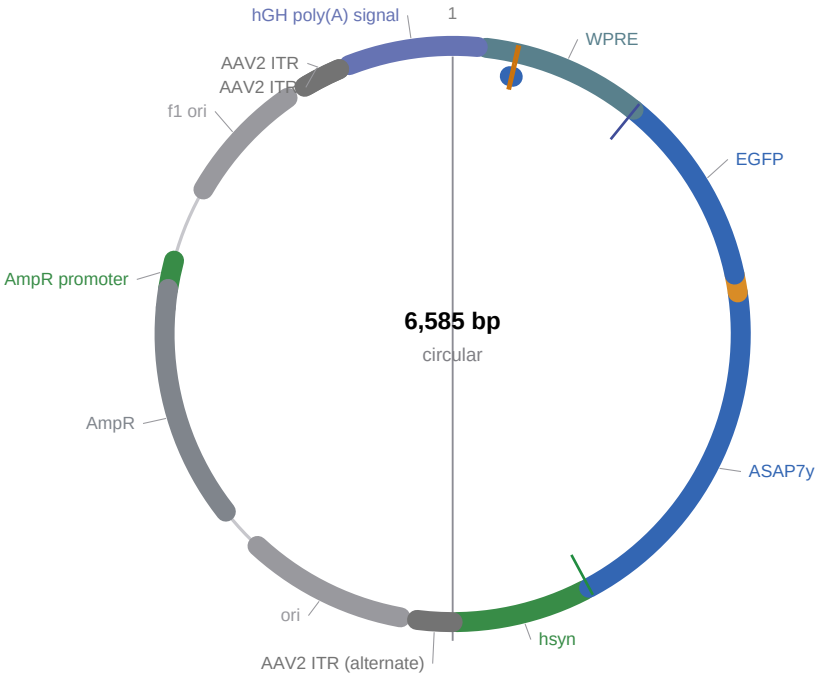
untitled

PCR + NEBuilder HiFi assembly

Final construct	6,585 bp, circular
Backbone	pBJ027_pAAV_hSyn_ASAP3_ (5,829 bp, Bill Jia)
Architecture	hsyn - ASAP7y - P2A - EGFP - WPRE - Factor Xa site - hGH poly(A) signal - sprBJ029 (HGH-pA REV) - AAV2 ITR - AAV2 ITR - f1 ori - AmpR promoter - AmpR - AmpR - ori - AAV2 ITR (alternate)
Workload	12 steps, 5 PCR(s), 1 assembly reaction(s)
Time	4.8 h hands-on, 2.8 days elapsed
Suggested name	pAAV-hSyn-ASAP7y-P2A-EGFP

Designed construct

untitled 6,585 bp



Outer ring reads forward, inner ring reverse. Position 1 is at the top.

- no stop codon at 3,052 in Factor Xa site
- no ATG (ATT) at 3,061 in Factor Xa site

Reagents and consumables

Quantities are per reaction. Check against the current package insert for the kits in use.

Item	Amount
5x high-fidelity reaction buffer	10 uL
10 mM dNTPs	1 uL
10 uM forward primer	2.5 uL
10 uM reverse primer	2.5 uL
template plasmid (1-10 ng)	1 uL
high-fidelity DNA polymerase	0.5 uL
nuclease-free water	to 50 uL
completed PCR reaction	all
DpnI	1 uL
agarose gel	0.8-1 % (1.5 % below 500 bp)
DNA stain	per kit

gel extraction kit	per kit
PCR or digest reaction	all
spin column or SPRI beads	per kit
2x assembly master mix	10 uL
linearised vector	50-100 ng
each insert	2:1 molar ratio over vector
competent cells	50 uL
assembly or ligation product	2 uL
SOC medium	950 uL
colonies	4-8
screening primer pair	per reaction
Taq master mix	per reaction
miniprep DNA	per facility spec
sequencing primers	one per junction

Primers to order

Lower case is the overlap or adapter tail; upper case binds the template. The quoted Tm is for the template-binding region only, which is what sets the annealing temperature.

Name	Dir	nt	Tm C	GC	Sequence (5' to 3')
backbone3-F	F	41	59.7	28%	cgagctgtacaagtaaAATCAACCTCTGGATTACAAAATTT
backbone3-R	R	18	59.8	56%	AGGAACCCCTAGTGATGG
hsyn-F	F	48	61.3	58%	tcactaggggttcctgcggccgcacgcgtGTGTCTAGACTGCAGAGGG
hsyn-R	R	18	63.2	56%	TTCTCGACTGCGCTCTCA
ASAP7y-F	F	51	60.5	56%	gagcgcagtcgagaaggtaccggatccgccaccATGGAGACGACTGTGAGG
ASAP7y-R	R	22	60.0	41%	GGTTACCACTTCAAGTTGTTTC
P2A-F	F	33	68.0	72%	cttgaagtggtaaccGGCTCCGGAGCCACGAAC
P2A-R	R	18	62.5	61%	GGGACCGGGGTTTTCTTC
EGFP-F	F	33	65.0	67%	gaaaaccccggtcccGTGAGCAAGGGCGAGGAG
EGFP-R	R	19	58.9	47%	TTACTTGTACAGCTCGTCC

Protocol

1. PCR Amplify the 4,004 bp backbone3 arc (1,955-130) from pBJ027_pAAV_hSyn_ASAP3_ with primers backbone3-F / backbone3-R

annealing 60 C; extension 2:00 for 4,004 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

2. PCR Amplify hsyn (485 bp) from pBJ027_pAAV_hSyn_ASAP3_ with primers hsyn-F / hsyn-R

annealing 61 C; extension 0:15 for 485 bp at 30 s/kb; 70% GC, add the high-GC enhancer; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase

to 50 uL	nuclease-free water

3. PCR Amplify ASAP7y (1281 bp) from paav-ef1a-dio-asap7y-wpre_ with primers ASAP7y-F / ASAP7y-R

annealing 60 C; extension 0:38 for 1,281 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

4. PCR Amplify P2A (66 bp) from pBJ230_pLJM1_JNKKTR_mK with primers P2A-F / P2A-R

annealing 62 C; primer Tm differ by 6 C, consider a gradient; extension 0:15 for 66 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

5. PCR Amplify EGFP (717 bp) from GUBD_opsin1 with primers EGFP-F / EGFP-R

annealing 59 C; primer Tm differ by 6 C, consider a gradient; extension 0:22 for 717 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

6. DPNI_DIGEST Digest the PCR reactions with DpnI to remove the parental plasmid templates (GUBD_opsin1, pBJ027_pAAV_hSyn_ASAP3_, pBJ230_pLJM1_JNKKTR_mK, paav-ef1a-dio-asap7y-wpre_)

37 C for 30-60 min, added directly to the PCR. Only cuts methylated (dam+) parental plasmid, not the amplicon.

7. GEL_EXTRACTION Gel purify hsyn (multiple products or a size-selection requirement)

Run far enough to resolve the bands, excise the correct band on a blue/long-wave transilluminator, elute in 20-30 uL.

8. DNA_CLEANUP Column purify the remaining PCR products

Elute in 20-30 uL. Quantify before assembly.

9. HI_FI_ASSEMBLY Assemble 5 fragments (hsyn + ASAP7y + P2A + EGFP + backbone3) using NEBuilder HiFi, 25 bp overlaps

50 C for 15 min (2-3 fragments) or 60 min (4 or more). Keep total fragment input near 0.03-0.2 pmol.

HiFi / Gibson assembly (20 uL)

10 uL	2x assembly master mix
50-100 ng	linearised vector

```
2:1 molar ratio over vector each insert
to 20 uL              nuclease-free water
```

10. `TRANSFORMATION` Transform the assembly into competent cells and plate on the appropriate antibiotic

Ice 30 min, 42 C for 30 s, ice 5 min, add SOC, recover 37 C for 1 h with shaking, plate on selective agar, incubate overnight at 37 C.

Transformation

```
50 uL                 competent cells
2 uL                  assembly or ligation product
950 uL                SOC medium
```

11. `COLONY_SCREEN` Screen 4-8 colonies by colony PCR, or diagnostic digest with NotI

Patch each colony before screening. Confirm candidates by diagnostic digest and set up overnight cultures.

12. `SEQUENCE_VERIFY` Sanger sequence across every junction, then confirm the full insert

Sequence across every junction first, then the full insert. Check the translated fusion, not only the DNA.

Detail

Everything behind the protocol above: how the design was validated, where each fragment comes from, what the other routes would cost, and the final sequence.

Validation

0 error(s), 7 warning(s), 5 informational.

Warnings

- dna: hsyn: 11 candidate parts matched; using 'hsyn'
- dna: P2A: 12 candidate parts matched; using 'P2A'
- dna: EGFP: the best-ranked 'EGFP' annotation (part 695, 717 bp) carries no stop codon and nothing translated follows it; used part 557 (717 bp) instead, which ends in one
- dna: EGFP: 12 candidate parts matched; using 'EGFP'
- dna: the backbone feature 'Kozak sequence' straddles an insertion point and was disrupted
- protein: fusion (Factor Xa site) does not begin with ATG
- protein: fusion (Factor Xa site) does not end in a stop codon

Checks

- protein: predicted fusion protein (ASAP7y + P2A + EGFP): 687 aa
- protein: predicted fusion protein (Factor Xa site): 4 aa
- dna: unique sites available for diagnostic digest: AgeI, ApaI, BamHI, BglII, BsrGI, ClaI, KpnI, MluI, NcoI, XbaI, XhoI
- packaging: carries the full AAV transfer plasmid element set
- verification: read back 4 of 4 parts from the assembled sequence, each once, in the specified order and orientation

Candidate parents

The request read as an ordered architecture: hsyn - ASAP7y - P2A - EGFP. The plasmid to start from is whichever already carries the longest leading run of it; the rest is covered by as few donors as possible.

Start from

Plasmid	Size	Owner	Already has
pBJ027_pAAV_hSyn_ASAP3_	5,829 bp	Bill Jia	hsyn (as hSyn promoter), ASAP7y (as Translation 648-1954)
paav-hsyn-dio-asap6c-kv-wpre_confirmed	6,331 bp	Byung Hun Lee	hsyn (as hSyn promoter), ASAP7y (as ASAP4c 413G)
YD007-pAAV-hSyn-CaMKAR	5,486 bp	Yangdong Wang	hsyn (as hSyn promoter)
Predicted_YA1406	5,539 bp	Yoav Adam	hsyn
YA1406_pAAV-hSyn	5,539 bp	Yoav Adam	hsyn
HFS24_c1maxi	5,599 bp	Felix Sigmund	hsyn (as hSyn promoter)

Then take the rest from 1 donor

- P2A - EGFP
- PCR_prdt2_p2A_Ch (1,820 bp, PZ) - has P2A, EGFP (as GFP)
- YQ0104 extraction (3,460 bp, Yitong Qi) - has P2A, EGFP (as GFP)
- YA1401_pENTR_Opt (5,956 bp, Yoav Adam) - has P2A, EGFP (as GFP)
- SLF169 (6,856 bp, Sami Farhi) - has P2A, EGFP (as GFP)

Cloning simulation

The primers below were amplified off the named templates and the products assembled, to check the steps build this construct rather than merely describe it.

Result: 6,585 bp product, 4/4 requested parts present - identical to the designed sequence

Part	Length	In the product
hsyn	485 bp	yes (100%)
ASAP7y	1,281 bp	yes (100%)

P2A	66 bp	yes (100%)
EGFP	717 bp	yes (100%)

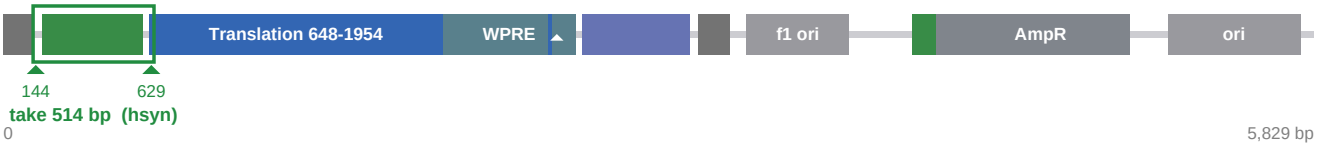
Where each fragment is cut from

The marked region is what the primers amplify. Each one is drawn on the plasmid it comes out of, in that plasmid's own coordinates.

pBJ027_pAAV_hSyn_ASAP3_ 5,829 bp - take 4,020 bp as backbone3

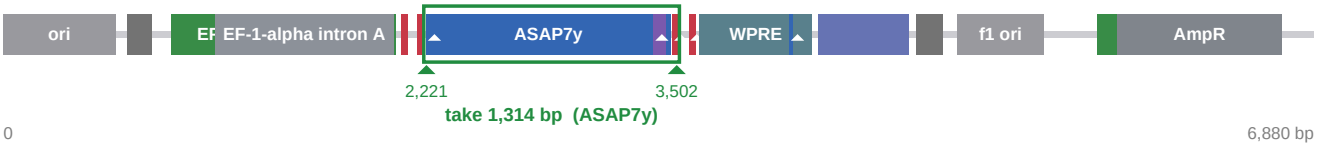
also: AAV2 ITR (alternate), hSyn promoter, Factor Xa site, hGH poly(A) signal, AAV2 ITR, AAV2 ITR, AmpR promoter, AmpR

pBJ027_pAAV_hSyn_ASAP3_ 5,829 bp - take 514 bp as hsyn



also: AAV2 ITR (alternate), hSyn promoter, Factor Xa site, hGH poly(A) signal, AAV2 ITR, AAV2 ITR, AmpR promoter, AmpR

paav-ef1a-dio-asap7y-wpre_ 6,880 bp - take 1,314 bp as ASAP7y

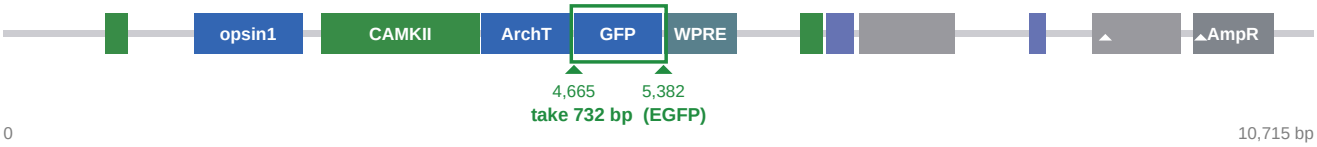


also: AAV2 ITR (alternate), lox2272, loxP, swap with Kir2.1 KGC, lox2272, loxP, Factor Xa site, hGH poly(A) signal, AAV2 ITR, AAV2 ITR

pBJ230_pLJM1_JNKKTR_mK 10,791 bp - take 81 bp as P2A

also: EF-1? promoter, EF-1? intron A (fragment), EF-1? intron A, P2A, GA linker, P (fragment), nef (fragment), HIV-1 pol (fragment), hPGK promoter, hPGK promoter

GUBD_opsin1 10,715 bp - take 732 bp as EGFP



also: 5'LTR, 3'LTR, BGH\polyA, F1\ori, SV40\Ori, SV40\Early\PolyA, pUC\Ori, ColE1 origin

How they go together

The simulated product, with each stretch named by the reaction it came out of.

assembled product 6,585 bp

also: P2A

Fragment	Cut from	Length	Lands at
backbone3	pBJ027_pAAV_hSyn_ASAP3_ 1955-130	4,020 bp	0-4,020
hsyn	pBJ027_pAAV_hSyn_ASAP3_ 144-629	514 bp	4,005-4,519
ASAP7y	paav-ef1a-dio-asap7y-wpre_ 2221-3502	1,314 bp	4,504-5,818
P2A	pBJ230_pLJM1_JNKKTR_mK 2128-2194	81 bp	5,803-5,884

EGFP	GUBD_opsin1	4665-5382	732 bp	5,869-16

Read-back

The assembled sequence re-annotated from scratch and compared with what was asked for, independently of the bookkeeping that built it.

read back 4 of 4 parts from the assembled sequence, each once, in the specified order and orientation

Route warnings

- template pBJ027_pAAV_hSyn_ASAP3_ is not sequence-verified
- template paav-ef1a-dio-asap7y-wpre_ is not sequence-verified
- P2A is only 66 bp; rather than a separate PCR, add it to an adjacent fragment's primer tail or order it as annealed oligos
- template pBJ230_pLJM1_JNKKTR_mK is not sequence-verified
- template GUBD_opsin1 is not sequence-verified
- junction 10 (AAV2 ITR|f1 ori): overlap arm is 76% GC; HiFi joining may be poor
- junction 11 (f1 ori|AmpR promoter): overlap arm is only 16% GC; overlap Tm will be low
- junction 12 (AmpR promoter|AmpR): overlap arm is only 24% GC; overlap Tm will be low
- primer backbone3-F: GC content 28% is low
- primer backbone3-F: 6 nt self-complementarity at the 3' end; primer-dimer risk
- primer backbone3-R: annealing region occurs 2 times in the template; expect mispriming
- primer hsyn-F: GC-rich 3' end (4+ of the last 5 bases); may misprime
- primer ASAP7y-F: possible hairpin with a 6 bp stem
- primer P2A-F: annealing Tm 68.0 C is well above the 60 C target
- primer P2A-F: GC content 72% is high
- primer EGFP-F: GC content 67% is high
- primer EGFP-R: GC-rich 3' end (4+ of the last 5 bases); may misprime

Fragments and sources

Fragment	Length	Source template	Synthesis
hsyn	485 bp	pBJ027_pAAV_hSyn_ASAP3_	no
ASAP7y	1,281 bp	paav-ef1a-dio-asap7y-wpre_	no
P2A	66 bp	pBJ230_pLJM1_JNKKTR_mK	no
EGFP	717 bp	GUBD_opsin1	no

Junctions

#	Left	Right	Frame shift	Notes
1	hsyn	ASAP7y	0	ok
2	ASAP7y	P2A	0	ok
3	P2A	EGFP	0	ok
4	EGFP	WPRE	0	ok
5	WPRE	Factor Xa site	0	ok
6	Factor Xa site	hGH poly(A) signal	0	ok
7	hGH poly(A) signal	sprBJ029 (HGH-pA REV)	0	ok
8	sprBJ029 (HGH-pA REV)	AAV2 ITR	0	ok
9	AAV2 ITR	AAV2 ITR	0	ok
10	AAV2 ITR	f1 ori	0	overlap arm is 76% GC; HiFi joining may be poor
11	f1 ori	AmpR promoter	0	overlap arm is only 16% GC; overlap Tm will be low
12	AmpR promoter	AmpR	0	overlap arm is only 24% GC; overlap Tm will be low
13	AmpR	AmpR	0	ok
14	AmpR	ori	0	ok
15	ori	AAV2 ITR (alternate)	0	ok

Route comparison

Route	Steps	PCR	Assembly	Hands-on	Elapsed	Cost	Usable
PCR + NEBuilder HiFi assembly (this one)	12	5	1	4.8 h	2.8 d	108	yes
Swap by digest + one PCR + HiFi assembly	0	0	0	0.0 h	0.0 d	1000	no
Swap by PCR + HiFi assembly (2 parents)	0	0	0	0.0 h	0.0 d	1000	no
Swap by restriction digest + ligation (2 parents)	0	0	0	0.0 h	0.0 d	1000	no
Restriction digest + ligation	0	0	0	0.0 h	0.0 d	1000	no

Swap by digest + one PCR + HiFi assembly was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, EGFP), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Swap by PCR + HiFi assembly (2 parents) was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, EGFP), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Swap by restriction digest + ligation (2 parents) was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, EGFP), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Restriction digest + ligation was not usable: the insert is made of 4 pieces from 4 donor plasmid(s); a ligation of that many fragments is not practical, so this construct needs the PCR/HiFi route (or a staged build of two pieces at a time)

Final sequence

6,585 bp, 54.0% GC, circular.

```
1  gcggccgcac gcgtgtgtct agactgcaga gggccctgcg tatgagtgca agtgggtttt
61  aggaccagga tgaggcgggg tgggggtgcc tacctgacga ccgaccccga cccactggac
121  aagcacccaa cccccattcc ccaaattgcg catcccctat cagagagggg gaggggaaac
181  aggatgcggc gaggcgcgtg cgcactgcca gcttcagcac cgcggacagt gccttcgccc
241  ccgcctggcg gcgcgcgcca ccgccgcctc agcactgaag gcgcgctgac gtcactcgcc
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361  gccggaccgc accacgcgag gcgcgagata ggggggcacg ggcgcgacca tctgcgctgc
421  ggcgccggcg actcagcgct gcctcagtct gcggtgggca gcggaggagt cgtgtcgtgc
481  ctgagagcgc agtcgagaag gtaccggatc cgccaccatg gagacgactg tgaggtatga
541  acaggggtca gagctcacta aaacttcgag ctctccaaca gcagatgagc ccacgataaa
601  gattgatgat ggtcgtgatg agggtaatga acaagacagc tgttccaata ccattaggag
661  aaaaatttcc ccgtttgtga tgtcatttgg attcagagta tttggagttg tgcttatcat
721  tgtagacatc atagtggtga ttgtggatct ggccatcagt gagaagaaaa gaggcattag
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841  gagagtgttt gttgaaggct tcaagaacta tttccggtcc aaactgaata ctttggatgc
901  agtcatagta gtgggcactc tgctaattgc tatgacctac tccttctctg ctcttgctgc
961  ctttaaatcc gagaacgtgt atattaccgc ggataaacag aaaaacggca ttaaagcgaa
1021  ctttaccgtg attcataacg tggaagatgg cagcgtgcag ctggcggatc attatcagca
1081  gaacaccccg attggcgatg gcccggtgct gctgccggat aaccattatc tgagctacca
1141  gaccgtgctg agcaaagatc cgaacgaaaa acgcgatcac atggtgctgc tggaagacgt
1201  gaccgcagcg ggcattacac acggcatgga tgaactgtat ggcggcaccg gcggcagcgc
1261  gagccagggc gaagaactgt ttaccggcgt ggtgccgatt ctggtggaac tggatggcga
1321  tgtgaacggc cataaattta gcgtgcgcgg cgaaggcgaa ggcgatgcga ccattggcaa
1381  actgaccctg aaatttattt gcaccaccgg caaactaccg gtgccgtggc cgaccctggt
1441  gaccacctta acctatggcg tgcagtgctt tagccgctat ccggatcata tgaaacgcca
1501  tgatttttt aaaagcgcga tgccggaagg ctatgtgcag gaacgcacca ttagctttaa
1561  agatgatggc aaatataaaa cccgcgcggt ggtgaaattt gaaggcgata ccctggtgaa
1621  ccgcattgaa ctgaaaggca ccgattttaa agaagatggc aacattctgg ggcataaact
1681  ggaatataac attgatcaga tgccgcagat ggttactctt cttcgagttc tgagaattgt
1741  tatcttaata agagtgattc gcctggcttc acagaagaaa caacttgaag tggtaaccgg
1801  ctccggagcc acgaacttct ctctgttaaa gcaagcagga gacgtggaag aaaaccccgg
1861  tcccgtgagc aagggcgagg agctgttcac cggggtggtg cccatcctgg tcgagctgga
1921  cggcgacgta aacggccaca agttcagcgt gtccggcgag ggcgagggcg atgccaccta
1981  cggcaagctg accctgaagt tcatttgcac caccggcaag ctgcccgtgc cctggcccac
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2101  gcagcacgac ttcttcaagt ccgccatgcc cgaaggctac gtccaggagc gcaccatctt
2161  cttcaaggac gacggcaact acaagacccg cgccgaggtg aagttcgagg gcgacaccct
2221  ggtgaaccgc atcgagctga agggcatcga cttcaaggag gacggcaaca tcctggggca
2281  caagctggag tacaactaca acagccacaa cgtctatatc atggccgaca agcagaagaa
2341  cggcatcaag gtgaacttca agatccgcca caacatcgag gacggcagcg tgcagctcgc
```

```
2401  cgaccactac cagcagaaca cccccatcgg cgacggcccc gtgctgctgc ccgacaacca
2461  ctacctgagc acccagtccg ccctgagcaa agaccccaac gagaagcgcg atcacatggt
2521  cctgctggag ttcgtgaccg ccgccgggat cactctcggc atggacgagc tgtacaagta
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3121  tcgccttcgc cctcagacga gtcggatctc cctttgggcc gcctccccgc atcgataccg
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3241  gccctggaag ttgccactcc agtgcccacc agccttgtcc taataaaatt aagttgcatc
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